Aditya Kumar Kushwaha

Software Engineer (Backend) | Node.js, Python (FastAPI), Go | AI-powered Systems

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GitHub Portfolio LinkedIn

Professional Summary

Software Engineer (Backend), 2 years of production experience in Node.js, Express.js, FastAPI and Go. Own two products at Spirehubs and run a live SaaS (PushPostVault) on self-managed infrastructure with Docker and Grafana. Strong in API design, RBAC/JWT auth, async processing and applied AI (RAG, agents, LLM integrations).

Experience

Spirehubs Softwares Aug 2025 – Present

Software Engineer (Backend & Applied AI) *Python, FastAPI, Node.js, Go, Redis, Docker*

- Own two products end to end: **DWIVE** (Caribbean ride-hailing platform) and **Spire Engage** (multi-client AI agent platform).
- DWIVE: split a FastAPI monolith into a modular structure, implemented RBAC across 100+ routes, migrated auth to a two-token (access + refresh) JWT system, and unified API response formats.
- Introduced queues, rate limiting and background jobs for reliability and throughput; built a Go-based AI report analysis system for contextual queries across reports with 1000+ references.

Evren Global Solutions Feb 2025 – Aug 2025

Full Stack Developer (MERN) *MongoDB, Express.js, React, Node.js*

- Developed **IBIEA2025**, managing 100+ award recipients, and **ICCIC**, supporting 200+ researchers in paper submission workflows.
- Implemented an AI-based visa document validation pipeline (data extraction, verification, vendor routing).

Projects

PushPostVault GitHub | Live

Go, Next.js, PostgreSQL, Redis, Cloudflare R2, Docker, Nginx, Grafana

- Built and run a live file-sharing SaaS (25 registered users, 4–5 active daily) with resumable chunked uploads, SHA-256 verification, atomic Redis-backed state machine with optimistic locking on PostgreSQL, and Razorpay billing.
- Self-hosted on a VPS with Docker and Nginx; centralized service logs and dashboards in Grafana, and files stored on Cloudflare R2.

AI PR Review Bot GitHub | Live

Node.js, Express, FastAPI, Neo4j, PostgreSQL, Inngest

- Built a multi-tenant GitHub App that reviews pull requests using a Neo4j code-dependency graph to flag cross-file breaking changes.
- Three decoupled services (webhook, Node/Express API, FastAPI AI engine); AES-256-GCM encrypted bring-your-own-key and inline @mention follow-up chat.

ArchadiLM GitHub | Live

Go, FastAPI, Qdrant, PostgreSQL, Redis Streams

- Built a NotebookLM-style RAG assistant supporting document, web and video ingestion with grounded, citation-backed responses streamed via SSE.
- Implemented HyDE query expansion, Reciprocal Rank Fusion and cross-encoder reranking; async ingestion through Redis Streams workers.

archadi-agent (Agent SDK) GitHub | npm

TypeScript, Node.js

- Built and published a TypeScript Agent SDK on npm: fluent builder API, function-calling runner, multi-agent handoffs, guardrails, structured output, streaming events, tracing, model fallback and a CLI.

Technical Skills

Languages: JavaScript, TypeScript, Python, Go
Backend: Node.js, Express.js, FastAPI, REST APIs, WebSockets, SSE, JWT, RBAC, Razorpay, Stripe, BullMQ, Celery
AI / LLM: LLM APIs, RAG, AI Agents, Function Calling, MCP, Prompt Engineering
Databases: PostgreSQL, MongoDB, MySQL, Redis, Qdrant, Neo4j
Infrastructure: Docker, Docker Compose, Linux, Nginx, VPS, Grafana, Cloudflare R2, GitHub Actions (CI/CD)
Frontend & Tools: React.js, Next.js, Git, GitHub

Education

Swami Vivekanand College of Science and Technology Bhopal

B.Tech in Computer Science and Engineering *2023 – 2026 | CGPA: 7.5*

Government Polytechnic Gopalganj

Diploma in Computer Science and Engineering *2019 – 2022 | CGPA: 8.5*

Achievements: 2nd position overall in the CS branch (Diploma); Student Head of Training and Placement.